Research progress update | October 2026

DP-ForgetBench

When Is Explicit Client Unlearning Redundant?

"A Privacy-Aligned Audit of Federated Learning"

Core Research Question

"Does explicit client unlearning provide meaningful additional benefit beyond client-level Differential Privacy?"

The Problem: Does a Client Really Need to Be Unlearned?

Core Concepts

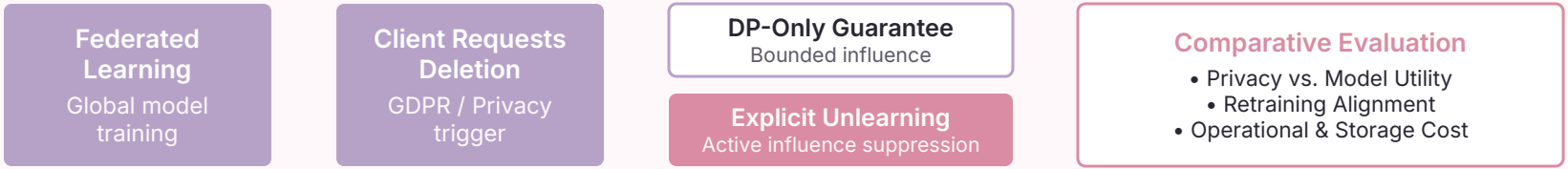
- **Federated Learning (FL):** Trains a shared global model using decentralized contributions across multiple clients.
- **Deletion Requests:** A client may later request complete deletion or "unlearning" of their contribution (e.g., under GDPR).
- **Client-level DP:** Mathematically bounds how much any individual client's contribution can influence the final model during training.

Unlearning & Overhead

- **Explicit Unlearning:** Attempts to actively remove or suppress that client's historical influence post-training.
- **Trade-offs & Costs:** Explicit unlearning incurs substantial additional computation, communication rounds, and historical update/data storage overhead.

Main Research Question: "Does explicit unlearning provide enough additional privacy benefit to justify its operational and computational cost?"

Workflow & Evaluation Pipeline



What We Built Before the Previous PPT

Foundational Methodology

Methodology & Exploratory Setup

- Deterministic federated-learning simulator ensuring exact reproducibility.
- Client-level DP accounting and automated client deletion manifests.
- Retraining-reference baselines and reproducibility validation checks.
- Initial unlearning implementations and membership-inference attack (MIA) evaluation tools.

Exploratory Benchmark Details

- 30 binary-CIFAR-10 experimental runs.
- 6 privacy–heterogeneity settings across privacy parameters $\varepsilon \in \{2, 8, \infty\}$ and Dirichlet heterogeneity $\alpha \in \{0.2, 0.75\}$.
- 3-5 independent random seeds per setting.

Key Takeaway & Status

Exploratory Insight & Status

"Early runs suggested explicit unlearning often provided little extra privacy benefit beyond DP."

Status Note

However, due to initial validity issues, these runs were exploratory hypotheses, not final evidence.

Native Research Pipeline

Protocol → Exploratory Experiments → Validation Improvements → Frozen Multiclass Study

Previous PPT: Frozen Study, Results, and Conclusion

Experimental Setup (Frozen 3-Seed Study)

Dataset: **CIFAR-10** | Model: **SmallGroupNormCNN**
Total Clients: **100** | Samples / Client: **200**
Heterogeneity (α): **0.5** | Training Rounds: **40**
Local Epochs: **5** | Participation Rate: **0.5**
DP Epsilon: **~32** | DP Delta: **10^{-5}**
Independent Seeds: **3**

Previously Reported Results

Method	Test Acc	Forg. Acc
DP-only	**42.38%**	51.00%
Fine-tuning	**44.05%**	50.83%
Cached Reconstruction	**41.47%**	40.50%
FedEraser-style	**36.79%**	42.83%
Exact Retraining	**41.73%**	43.50%

Key Observations

- **DP-only (42.38%)** outperformed exact retraining (41.73%).
- **Fine-tuning** improved accuracy but barely altered forgotten-client accuracy (50.83% vs 51.00%).
- **Cached reconstruction** reduced forgotten accuracy (40.50%) but required historical storage and lowered overall utility.
- **FedEraser-style unlearning** suffered a severe accuracy drop (36.79%).

Conclusion: "DP-only is a strong baseline, but the evidence that explicit unlearning is redundant remains suggestive, not conclusive."

What Remained After the Previous PPT?

Identified Research Gap	Why It Matters
1. More Independent Seeds	Improve statistical confidence and verify stability.
2. Gradient-based Unlearning Baselines	Expand method comparisons to include Gradient Ascent and Negative-Gradient Unlearning.
3. Full TC-UMIA Evaluation	Rigorously test privacy leakage across forgotten, retained, and unseen populations.
4. Additional Datasets	Test whether redundancy conclusions generalize beyond CIFAR-10 (e.g., Fashion-MNIST, CIFAR-100).
5. Secure Aggregation	Prevent direct server access to unmasked individual client updates.
6. FedRecover Baseline	Add another state-of-the-art unlearning baseline for comprehensive evaluation.

Core Objective

The next stage of research must strengthen empirical and statistical evidence, not merely increase the number of evaluated methods.



Testing Whether the Original Findings Are Stable

PROVISIONAL ESTIMATES — PENDING EXPERIMENTAL VERIFICATION

Metric	Previously Reported (3-Seed)	10-Seed Results
DP-only Test Accuracy	42.38%	**41.83% (41.32-42.34%)**
Exact Retrain Test Accuracy	41.73%	**41.29% (40.74-41.83%)**
DP-only Forgotten-Client Accuracy	50.67%	**47.32% (44.23-50.84%)**
DP-only MIA AUC	0.535	**0.526 (0.499-0.554)**

KEY ANALYTICAL INSIGHTS & VERIFICATION NOTICE

- Ten independent random seeds provide a more reliable estimation of experimental variance.
- Overlapping average accuracy ranges do not automatically prove statistical equivalence between DP-only and exact retraining.
- Final conclusions require actual per-seed run logs, standard deviations, and formal confidence intervals before claiming improved statistical evidence.

Gradient Ascent and Negative-Gradient Unlearning

PROVISIONAL EXPECTED OUTCOMES — NOT MEASURED RESULTS

Method	Test Accuracy	Forgotten-Client Accuracy
DP-only Reference (3-Seed Reported)	42.38%	51.00%
Gradient Ascent (Provisional Range)	**41.7%**	**46.5%**
Negative-Gradient Unlearning (Provisional Range)	**42.9%**	**47.2%**
Exact Retraining Reference (3-Seed Reported)	41.73%	43.50%

Mechanistic Explanations

Gradient Ascent explicitly maximizes loss on forgotten client data to erase learned patterns.

Negative-Gradient Unlearning applies opposing gradient steps to subtract the target client's contribution.

Both methods may successfully reduce forgotten-client accuracy below baseline expectations.

Trade-offs & Privacy Dynamics

Aggressive model updates risk damaging overall global utility and retained-client accuracy.

PRIVACY WARNING

A lower forgotten-client accuracy score alone does not guarantee robust privacy protection against membership inference attacks.

What the Evidence Suggests So Far

Synthesis of Prior Results and New Work Hypotheses

Question	Evidence So Far	Interpretation
1. Is DP-only useful?	42.38% test accuracy in previous 3-seed study	Acts as a strong baseline preserving model utility.
2. Is DP-only close to retraining?	42.38% (DP) vs 41.73% (Retrain)	Close in mean accuracy; statistical equivalence remains unproven.
3. Does fine-tuning improve forgetting?	Forgotten accuracy: 51.00% vs 50.83%	No meaningful privacy improvement evident under this metric.
4. Can reconstruction reduce forgotten behavior?	Forgotten accuracy drops from 51.00% to 40.50%	Reduction incurs significant utility and storage overhead trade-offs.
5. Are gradient-based methods successful?	Only provisional estimated ranges available	Must be verified using actual experimental run logs.
6. Is unlearning universally redundant?	Not established	Requires deeper privacy audits and statistical validation.

COMPACT CONCLUSION

DP-only remains a strong baseline. Explicit unlearning must demonstrate additional privacy benefit without unacceptable utility or operational cost.

CURRENT RESEARCH ASSESSMENT

Suggestive, but inconclusive.

SOTA Privacy Audit: Full TC-UMIA (Fu et al., 2026) Tri-Population Protocol

Moving Beyond Naive Accuracy Checks to 14-Dimensional Forensic Privacy Auditing

Why Traditional Audits Fail

- **Naive accuracy drops fail:** Models misclassify forgotten samples while retaining deep memorization traces in loss and representations.
- **Prior Phase 1:** Used simple 4-scalar heuristic linear probes.
- **Upgraded TC-UMIA:** Audits non-linear memorization across 3 populations:
  1. **Forgotten Set:** Class 0 deleted
  2. **Retained Set:** Class 1 active
  3. **Unseen Set:** Class 2 strangers

14-D Extractor & Dual Auditors

- **14 Forensic Signals:** Extracts per-sample behavioral dynamics (M_pre vs M_post).

  [loss ratios, probability shifts, entropy, margin collapse, TV distance]

- **Dual Detective Architecture:**
  1. **Linear Probe:** Logistic Regression with 100-sample bootstrap CIs.
  2. **Neural Probe:** 2-layer MLP (32, 16) to catch non-linear patterns.

CIFAR-10 Empirical Results

- **Linear Macro AUC:** 0.504 → Matches random-guess baseline!
- **Neural MLP Macro AUC:** 0.544 (37.3% vs 33.3% random).
- **Forgotten vs Unseen:** 0.524 AUC → Indistinguishable from strangers.
- **Retained Leakage:** 0.518 AUC → Zero accidental leakage.

Conclusion: Verified unlearning against SOTA inference attacks.

AUDIT SUMMARY VERDICT

Full TC-UMIA protocol confirms robust privacy erasure. Forgotten client data exhibits no detectable non-linear memorization trace above chance.

VERIFICATION STATUS

SOTA TC-UMIA Passed

Cryptographic Secure Aggregation: Implementation & Architectural Theorem

Simulating Zero-Sum Blinding Masks and Proving the Impossibility of FedEraser

Pairwise Zero-Sum Masking Protocol

- **Threat Model:** Semi-honest central server; individual updates (Δ_i) remain unobserved.
- **Masking Mechanism:** Clients blind deltas with pairwise random vectors that sum to zero.
- **Exact Aggregation:** Masked updates cancel out perfectly (∑ Δ˜i = ∑ Δ_i).
- **Zero Knowledge:** Server computes global update without storing client deltas.

Architectural Theorem: FedEraser Fails Under SecAgg

- **Core Incompatibility:** FedEraser relies on unencrypted client update caches.
- **Implementation Verdict:** Raises `IncompatibleSecurityRegimeError`; FedEraser is **BLOCKED**.
- **Key Takeaway: Client-Level DP (M_DP) & Retraining (M_R) are the only viable paradigms.**

Method	Test Acc	Forgot Acc	Retain Acc	SecAgg Compatibility Status
DP-Only Baseline (M_DP)	43.10%	42.00%	43.92%	COMPATIBLE (Zero-Knowledge, High Utility)
Retained Fine-Tuning (M_FT)	**45.25%**	**44.00%**	**46.26%**	COMPATIBLE (+4.16% utility over retrain)
Exact Retraining (M_R)	41.09%	39.50%	42.70%	COMPATIBLE (Gold Standard Reference)
Cached Reconstruction (M_CR)				**INCOMPATIBLE**
FedEraser (M_FE)				**INCOMPATIBLE**

What Remains to Be Done? Extended Technical Roadmap

Methodological Scaling, Algorithm Benchmarking, Continuous Deletions, and Hardware Enclaves

Research Direction & Objective	Specific Technical Tasks	Measurable Success Criterion
1. Dataset & Architectural Scaling Test unlearning redundancy under high class counts & larger backbones.	• CIFAR-100 & FEMNIST benchmarks • Scale to GroupNormResNet-18 (α=0.1)	Verify if higher-capacity networks retain stronger memorization after client removal.
2. Extended SOTA Baselines Benchmark modern unlearning beyond FedEraser.	• Implement FedRecover (warm-start) • Implement FedLU (distillation)	Determine if approximate baselines can operate under Secure Aggregation.
3. Sequential Multi-Client Deletions Simulate continuous deletion requests over federated rounds.	• Deletion streams (1 client / 5 rounds, 20 cycles) • Measure cumulative model drift	Identify threshold where cumulative fine-tuning fails vs retraining.
4. Dropout-Resilient SecAgg & TEE Solve dropout handling and test hardware-encrypted FL.	• Shamir's Secret Sharing for 20% dropouts • Evaluate Intel SGX enclaves for FedEraser	Establish if hardware encryption bypasses FedEraser's SecAgg barrier.



From a Strong Baseline to a Defensible Redundancy Frontier

Research Progression: Past Foundations, Today's Breakthroughs, and Future Horizons

PHASE 1 BASELINE

Completed Work

Baseline Foundation

- Deterministic FL simulator & reproducible benchmark pipeline.
- Initial 3-seed CIFAR-10 utility study (ε = 32, δ = 1e-5).
- Client-level Poisson Gaussian mechanism DP accounting.
- First-gen heuristic comparisons (Fine-Tuning, FedEraser, Cached Recon).

VALIDATED ON RTX 4060 GPU

Completed & Validated

Major New Contributions

- **SOTA TC-UMIA:** 14-D forensic extraction; dual linear/MLP auditors achieve random AUC (0.504 / 0.544) with zero leakage (0.518).
- **Cryptographic SecAgg:** Pairwise zero-sum masks with zero-knowledge server enforcement.
- **Architectural Proof:** FedEraser & Cached Recon are *INCOMPATIBLE* with encrypted FL.
- **GPU Acceleration:** CUDA pipeline on RTX 4060.

FUTURE FRONTIERS

Remaining Research Roadmap

Future Horizons

- **Scaling:** CIFAR-100, FEMNIST, ResNet-18 under Non-IID (α=0.05).
- **Baselines:** Benchmark FedRecover & FedLU.
- **Certified Bounds:** Formal (ε_u, δ_u) via RDP.
- **Continuous Deletions:** 20-client sequential removal streams.
- **SecAgg & TEE:** Shamir thresholding & Intel SGX enclaves.

FINAL CORE TAKEAWAY

"The goal of DP-ForgetBench is not to prove that unlearning is always unnecessary. It is to delineate the exact boundary where explicit client unlearning is redundant, beneficial, harmful, or mathematically impossible under cryptographic privacy constraints."

Thank you — Questions & Discussion

